

A composite medical image featuring a stethoscope, a human skeleton, a heart, and a DNA helix, all set against a blue background.

Faster and more accurate Skin Cancer Identification

By Victoria Vicheva 233182

Project Pitch

Business Understanding

First idea:

Faster and more accurate diagnosis of skin cancer, type Melanoma and other skin types of skin lesions

Main stakeholder:

Medical organizations such as hospitals and clinics

Application idea:

Diagnosis in a snap

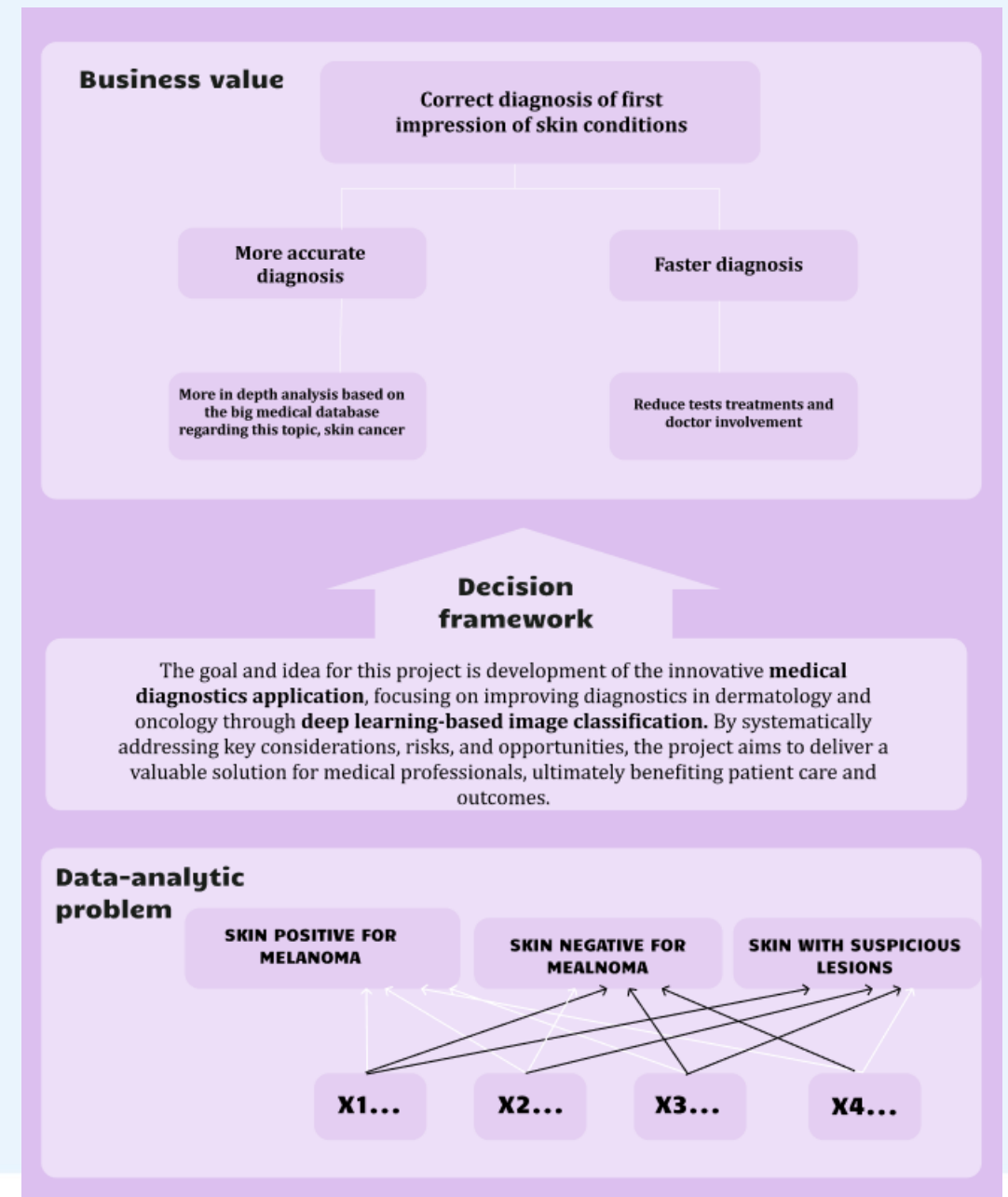


Fig.1: DAPS Diagram

Problem Overview

Deep Learning

What will this model predict?

Successful identification of skin cancer, type Melanoma.

How many classes are in the dataset?

“Skin with benign(non-cancerous) moles”

“Skin with suspicious lesions”

“Skin with Melanoma”

In each class:

200 images, in total there are 600 image in the dataset

Accuracy of Baselines:

Random Guess: around 33%

Human-Level Performance: 30-40%

Basic MLP: around 70%

Fig.1: Benign mole



Fig.2: Cherry angioma



Fig.3: Melanoma



Model Overview

Deep Learning

The process of creating and training the model:

Iteration 1: manually preprocessing + one-hot encoding

Iteration 2: Resizing + one-hot encoding

Iteration 3: Data Augmentation

Iteration 4: Transfer Learning, Xception model + Data Augmentation

Iteration 5: Transfer Learning, MobileNet

```
base_model5 = keras.applications.MobileNet(  
    include_top=False,  
    weights='imagenet',  
    input_shape=(128, 128, 3))  
  
# freezing the weights of the convolutional base  
base_model5.trainable = False  
  
extended_model5 = Sequential()  
# adding the base model to the extended model  
extended_model5.add(base_model5)  
extended_model5.add(Flatten()) # Flattening the output of MobileNet  
extended_model5.add(Dense(256, activation='relu'))  
extended_model5.add(Dense(128, activation='relu'))  
extended_model5.add(Dropout(rate=0.5))  
extended_model5.add(Dense(3, activation='softmax'))  
extended_model5.summary()  
  
# Preprocess of the images and get the outputs from the base model  
preprocessed_images_train5 = keras.applications.mobilenet.preprocess_input(v5_X_train)  
preprocessed_images_val5 = keras.applications.mobilenet.preprocess_input(v5_X_val)  
preprocessed_images_test5 = keras.applications.mobilenet.preprocess_input(v5_X_test)
```


Model Performance

Deep Learning

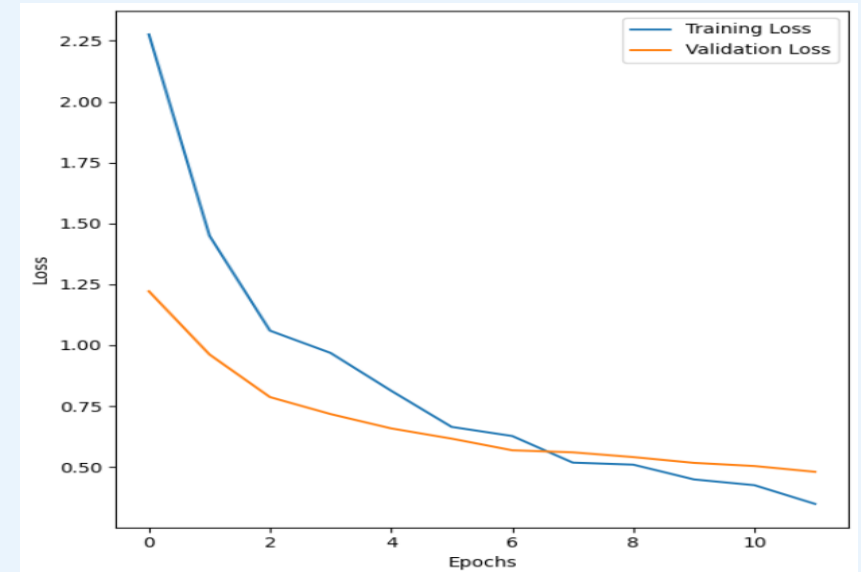
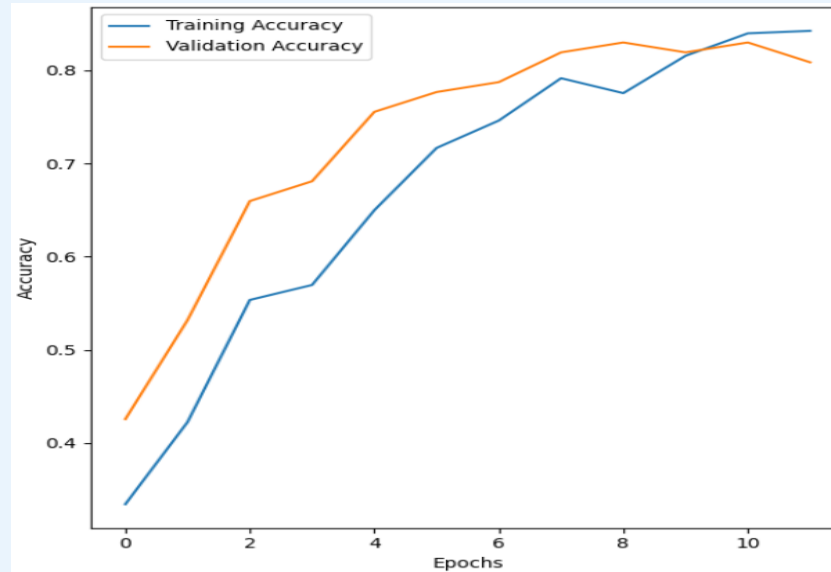
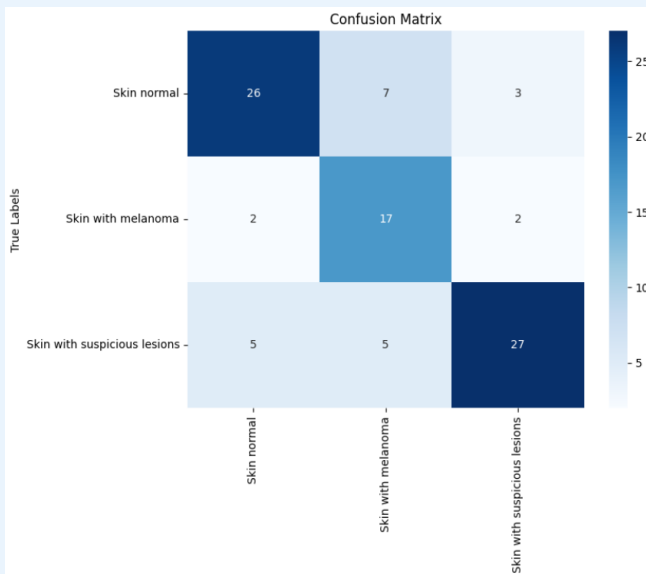
Results obtained with the final version of the model:

Final version:

Iteration5 Transfer Learning, MobileNet

75% Test Accuracy

Precision: 0.7648162892683251
Recall: 0.7446808510638298
F1 Score: 0.7485846438482886



Demo

Human-Centered AI

Some results from the error analysis



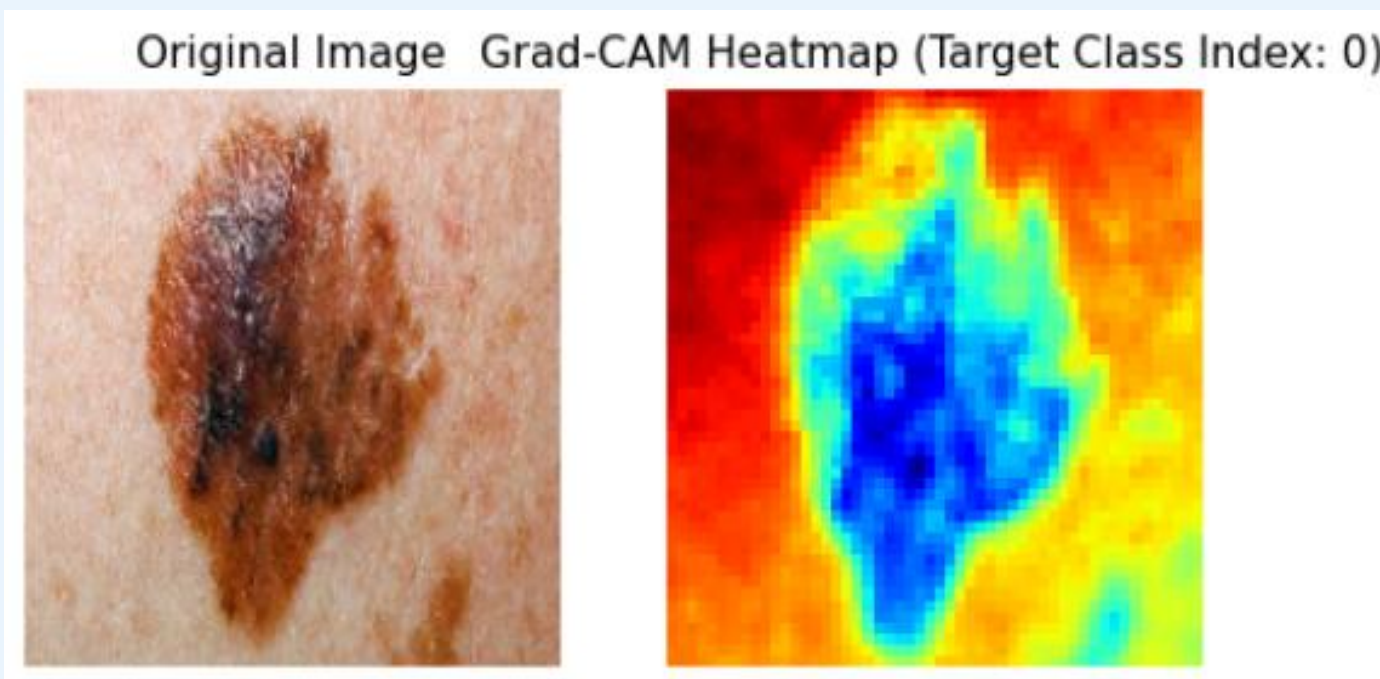
Model Interpretability

Responsible AI (1)

In the context of a medical image classifier for skin lesion detection, interpretability holds greater importance.

Example XAI methods:

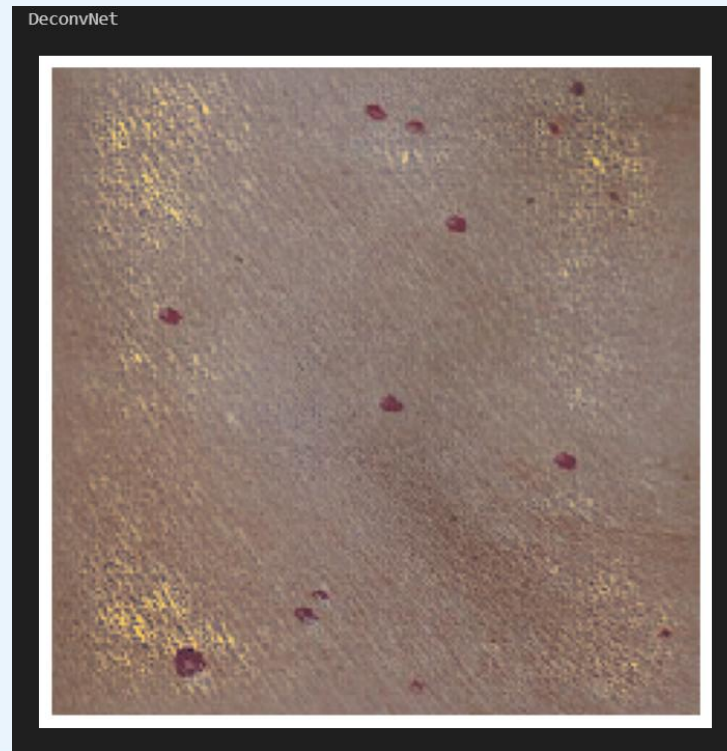
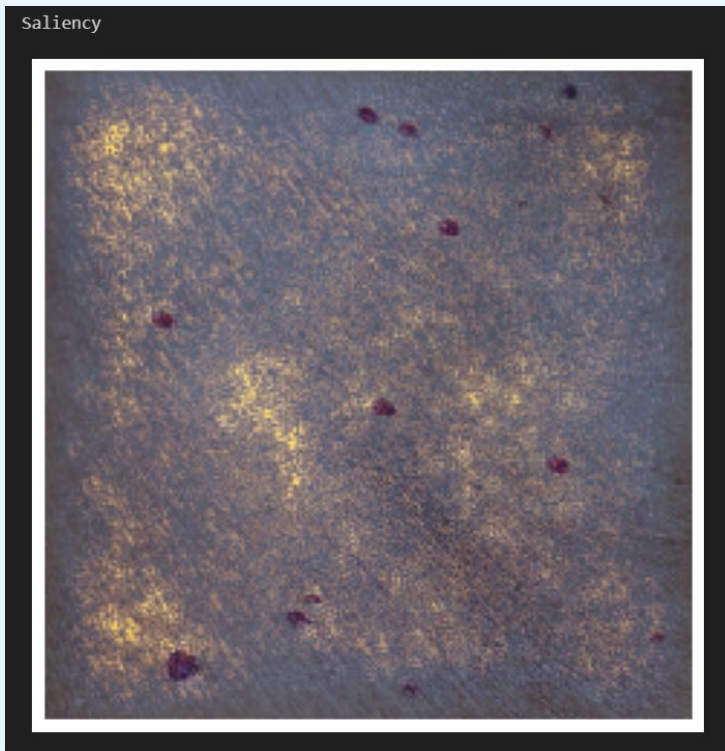
- Saliency
- DeconcNet
- Gradient Input
- Guided Backpropagation
- Grad-CAM



Fig

- Responsible AI(2)

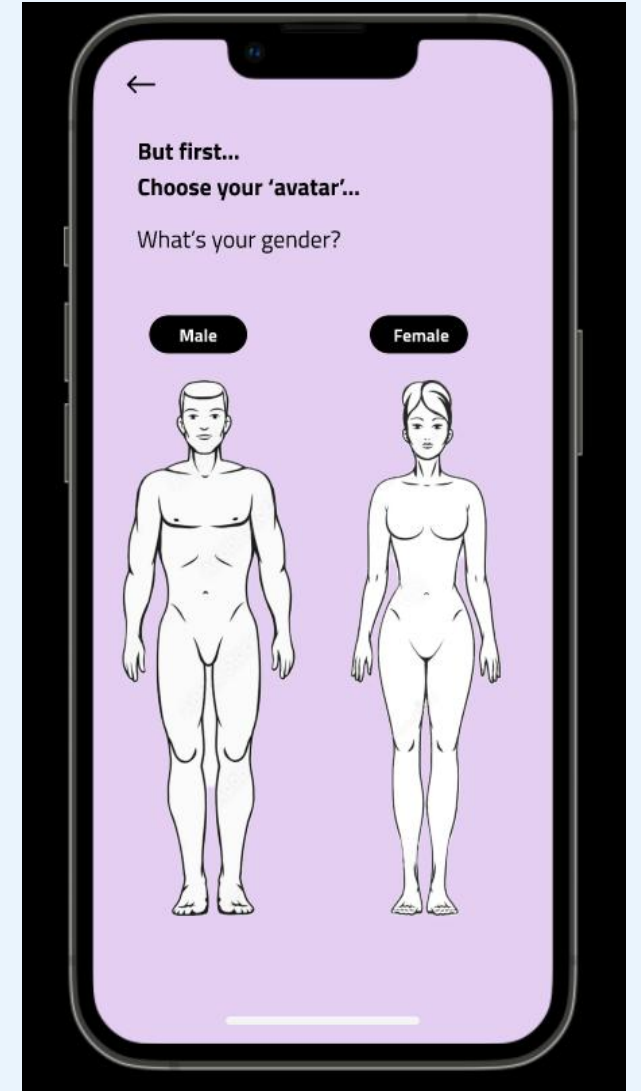
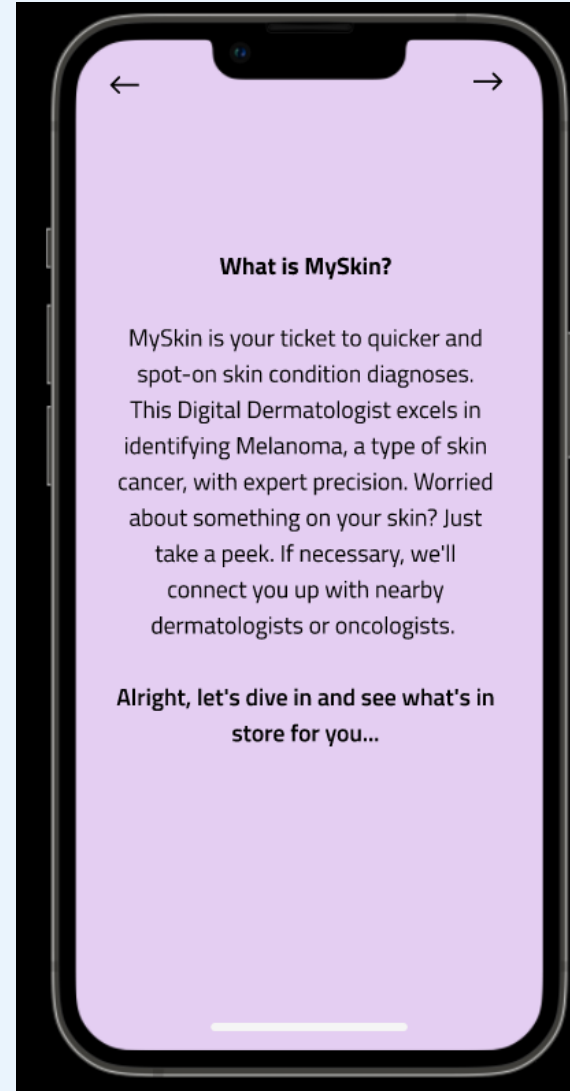
Other XAI methods



User Study

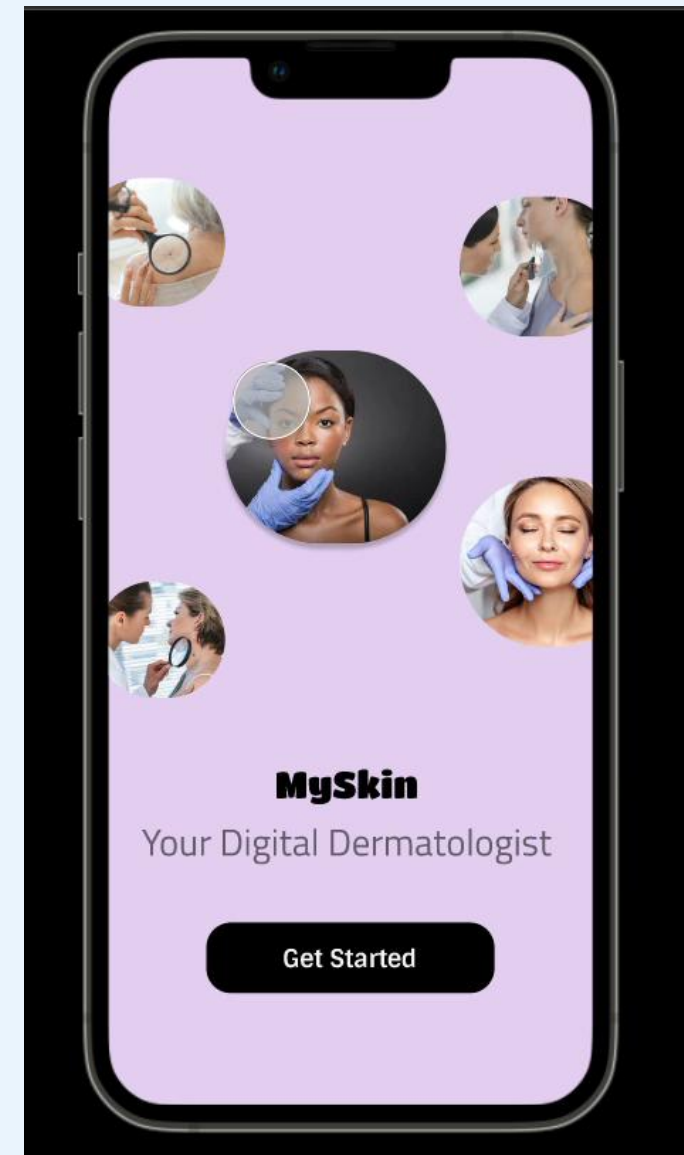
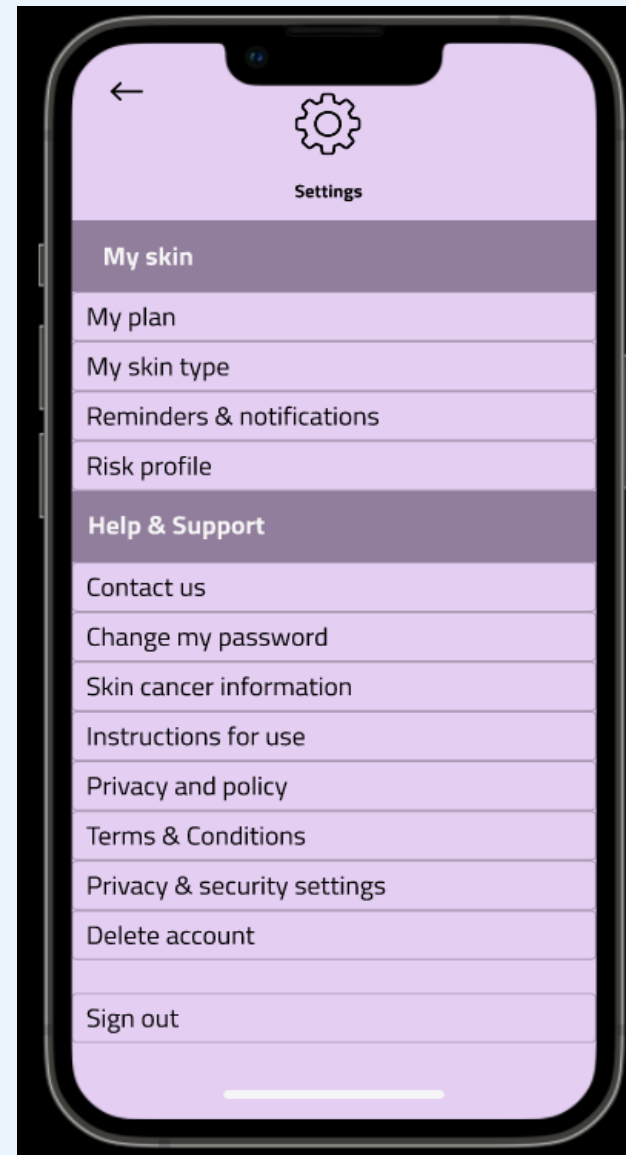
Human-Centered AI

- **Results of think-aloud study:**
 - Mostly positive
 - Users knew what to do
 - Confusion with the buttons
- **A/B test:**
 - Both positive and negative feedback
 - Most of the testers did not understand the exact purpose of the application



Demo

Human-Centered AI



Thank you for your attention!

Any questions?